

VB30

An aggressive diode-bridge programme limiter with capacitor memory, freeze and a gate.



Overview

VB30 is a characterful, aggressive limiter in the spirit of early broadcast modules and high-ratio industrial limiters. It is made for room mics, shouty vocals and parallel drum buses - the places where you want to hear the compression.

The signal runs gate, then limiter, then make-up gain, then MIX. The limiter has a fast, peak-sensing detector and a diode-bridge style gain cell that thickens as it works harder. Its release has a "capacitor memory": short peaks snap straight back, while sustained squashing clamps down and lets go slowly with a long, musical suck-back.

FREEZE stops the noise floor and the room from pumping up in the gaps: when the audio drops into a pause, the gain reduction is held where it is until the music returns. A gate or gentle expander sits in front of the limiter for tidying up spill.

Reach for it for

- Smashed room and ambience mics
- Parallel drum crush (with MIX)
- Shouty, in-your-face vocals
- Gritty snares and bass with the diode-bridge COLOUR
- Big pumping and suck-back effects

Controls

LIMITER (upper row) - DYNAMICS

THRESHOLD (orange plate)	The level where limiting starts, from -40 to 0 dB (-18 dB by default). Turn it down for more squash.
RATIO	From 1.5:1 up to LIMIT (fully right), 4:1 by default. The knee is soft.
ATTACK	Very fast, FET-style: from 50 microseconds to 20 ms (0.8 ms by default). Slower settings let more of each hit's front edge through.
RELEASE	From 30 to 1500 ms (200 ms by default). With MEMORY above zero this is the centre of a release that stretches and shrinks with the music.

LIMITER - CHARACTER

MEMORY	The capacitor memory, 0 to 100% (50% by default). The release follows how much gain reduction there has been over the last half second or so: after short peaks it recovers as fast as a quarter of RELEASE, after sustained squashing as slowly as six times RELEASE. 0 gives a plain, fixed release.
COLOUR	The diode-bridge gain cell, 0 to 100% (40% by default): a soft, slightly lopsided saturation that thickens as the gain reduction deepens. Its level is compensated, so it adds character rather than volume. 0 is clean.
GAIN	Make-up gain after the limiter, from -6 to +24 dB.

LIMITER - keys and meter

IN	Switches the limiter in or out. The POWER light in the header shows it.
LINK	On (the default): both sides share one gain, for the limiter and the gate, so the stereo image stays put. Off: each side

	works on its own.
MEMORY CHARGE	Twelve lights showing how much the memory is stretching the release right now, from SNAP (quick recovery) to SUCK (long, slow let-go).

GATE (lower row) - GATE / FREEZE

FREEZE THR (orange plate)	The level below which FREEZE holds the gain reduction, from -70 to -25 dB (-50 dB by default). A fast detector watches the input; there is 3 dB of hysteresis, so it does not chatter at the threshold.
GATE THR (orange plate)	The gate's threshold, from -80 to -20 dB (-60 dB by default).
RANGE	How far the gate turns the signal down, from 0 to 60 dB (15 dB by default).
RELEASE	How fast the gate closes, from 20 to 2000 ms (250 ms by default). It opens in about half a millisecond.

GATE - SIDECHAIN / MIX

SC HPF	A high-pass filter on what the limiter listens to, from OFF (fully left) up to 300 Hz. Raise it so the bass doesn't make everything pump; the bass itself isn't filtered.
MIX	Parallel blend with the dry signal, shown as WET 0 - 100. 100 is fully limited; lower settings tuck the crushed sound under the dry one.

GATE - keys and lamps

GATE / EXP	GATE: below the threshold the signal drops by the full RANGE. EXP (the default): a gentle 1:2 expander - the further below the threshold, the more it is turned down, up to RANGE.
OPEN	Lights while the gate is open.

FREEZE	Switches the freeze on or off (on by default). When the signal falls below FREEZE THR - a pause, the gap between hits - the limiter's release stops and the gain reduction stays where it is, so the noise floor and the room do not swell up in the gaps.
GATE IN	Switches the gate in or out (out by default). The readout underneath shows how much the gate is turning down, or GATE OUT.
HOLD	Lights while the freeze is holding the gain reduction.

Meters

GAIN RED.	Two glowing bars, L and R, showing the limiter's gain reduction from 0 to 30 dB, with the current figure underneath.
OUTPUT	Two LED columns, L and R, showing the output level from -36 to +3 dBFS.

The header

PRESETS	Opens the list of factory presets (a click on the display does the same); the ◀ / ▶ keys step through them. Choosing one sets the limiter and gate controls for that job. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Room Mic Smash	Limiting at -24 dB with a 0.2 ms attack, strong memory and colour, freeze on, SC HPF at 90 Hz.
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Parallel Drum Crush	12:1 at -28 dB with heavy memory and +12 dB of make-up, blended in at 40%.
Shouty Vocal	6:1 at -20 dB with moderate colour; the gate in as an expander to tidy the gaps.
Big Suck-back	8:1 with MEMORY at 100% and a 300 ms release for long, pumping let-go; freeze off.
Brick Wall Peaks	Limiting at -14 dB with the fastest attack and an 80 ms release: peak control with light colour.
Bass Grab	3:1 at -16 dB with a slower 4 ms attack and SC HPF at 60 Hz.
Snare Grit (gated)	10:1 with COLOUR at 85%, a hard gate (30 dB range) in front, SC HPF at 150 Hz, LINK off.
Gentle Glue Pump	4:1 at -18 dB with a 400 ms release and light colour.
Destroy (blend in)	Limiting at -40 dB, full colour, +18 dB of make-up, blended in at 30%.
Colour Only	1.5:1 with a slow attack and release and no memory - mostly the diode-bridge colour, little compression.
Reset all to defaults	Every control back to its starting value.

Tips

- For a room mic, keep FREEZE on and set FREEZE THR just above the room's noise between hits: the HOLD lamp should light in the gaps, not during the drums.
- Turn MEMORY up for the classic long suck-back on sustained material, or down to 0 when you want the release to stay predictable.
- On a parallel bus, crush hard with a low THRESHOLD and plenty of GAIN, then bring MIX down until the drums thicken without losing their attack.
- If kick or bass makes the whole signal duck, raise SC HPF to 90 - 150 Hz.

File: gfx-vb30.jsfx